FMI	ADMINISTRATIVE INFORMATION	FMI No: 25338

FMI Title: LightSpeed 4.X GRE & LinuxPegasus M4 Software

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☑ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☑ Staged FMI		**Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
	☑	**Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
	☐	**Parts Ordering**: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
	☐	**No Parts** are required to complete this FMI.
☐ Batch with		
		To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged		**No Customer List is provided**. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	**GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70**	**Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905**

Special Instructions:

Unit Tracked: 2366438, 2377708

FMI Instructions Part No.: 2402136-100

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25338 **FMI Title:** LightSpeed 4.X GRE & LinuxPegasus M4 Software

Customer Full Name: ______________________

Number and Street Address: ______________________

City and Country: ______________________

____

Country Code Site Number System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

____

____

Installer Comments ______________________

Effectivity

This FMI upgrades the software on CT host computer consoles at sites with any one of the following:

- 2366438, LightSpeed 4X 16Slice (H16) LinuxPegasus/GOC2 host computer console assembly
- 2377708, LightSpeed 4X 16Slice (H16) GRE/GOC3 host computer console assembly

NOTICE: **Do not execute this FMI / load this software on HighSpeed QX/I, LightSpeed 5.X Pro16, LightSpeed 5.X RT, or PET Discovery systems.**

Purpose

The purpose of this FMI is to upgrade the PC computer software on GOC2 (Linux/Pegasus) and GOC3 (GRE) consoles. GOC2 software is updated from its original M3 release to M4, where M3 dates back to March 2003. GOC3 software is updated from M3.1 to M4, where M3.1 was delivered by FMI 25330 in Nov 2003.

Linux/GOC2 customers get all s/w improvements since its original software release, 2369631, distributed in March 2003.

GRE/GOC3 customers get an incremental s/w improvement since GRE M3.1 FMI 25330.

This release has the following improvements, new features, and product safety fixes:

Improvements

- SmartPrep timing issues resolved
- Images reconstructed out of order
- Cardiac images that fail to reconstruct when there is a big jump in heart rate
- FastCal runs 16 x 0.625 detector
- No scan aborts due scan hardware hang
- Serial Over LAN CD/tool to change DARC IP address to prevent hanging hospital network (change DARC IP address 172.16.xxx if already used by hospital's network)
- Show Localizer lines no longer disappear
- Color save images cause network to crash
- Images rotated

New Features (LinuxPegasus / GOC2 Only)

- AutomA Changes
- Color Image
- RCOC – Remote Console Observation
- Common User Interface with new colors for CT
- Multiple Repeat Series
- 350ms Film composer deposit time
- Anonymous Scan Data Save/Restore
- Driver support for new Motorola MVME2400 single board computer (LightSpeed 4X (H16) only)

Product Safety Fixes (LinuxPegasus GOC2 & GRE GOC3)

- PSR 13009962 - H4.1M3: Hardware stopped and scan aborted.
- PSR 13010564 - H4.1M3: Gantry display went blank.
- PSR 13010982 - H3.1M4: Smart step images out of sequence.

Furnished Materials

Qty	Description --FMI Kit 25338, 2402680--	Part #
1	FMI Document	2402136-100
1	Linux Software Installation Procedure	2362590-100
1	GRE Software Installation Procedure	2378261-100
1	LightSpeed 4.X Technical Reference Manual	2351785-100
1	CT LightSpeed Series CD	2369740-247
1	LightSpeed Series Applications & Tips	2370773-100
1	GRE Service Information CD	2384451-200
1	LightSpeed 4.X (H16) GRE/GOC3 & LinuxPegasus/GOC2 M4 S/W Collector w/ Product Locator	5111411

Tool Required

Qty	Description	Part #
1	LightSpeed 4.X / 3.X / HS QXi Class C Software, M4 (Note: Part of “6-Pack” 2401759 Class C & M collector)	2394897
1	LightSpeed 4.X / 3.X / HS QXi Class M Software, M4 (Note: Part of “6-Pack” 2401759 Class C & M collector)	2394898

People Power

Travel: 1 Hour

Labor: 3 Hours

Steering Guide for Linux Pegasus (GOC2)

Prepare System For Upgrade

- ❑ Software Revision Check
- ❑ Archive Image Data
- ❑ System Operation Verification
- ❑ Check Software Loading Devices
- ❑ Save System State & CSA Protocols
- ❑ Record Key System Information
- ❑ Network Configuration Info
- ❑ HIS/RIS Configuration Info (Connect Pro)
- ❑ Insite Connectivity Info
- ❑ BroadBand Insite Connected Systems
- ❑ Installed Options
- ❑ Record Latest Gantry Revolution Entry

FMI 25338 Section 1
Direction 2362590-100

NOTE: Avoid re-loading OS software to enable other-than-English keyboard.

Be sure to select the correct language keyboard in the initial OS load process. This is the only method available to set the system to recognize an "other-than-English" keyboard. If you choose to install an international language keybboard at a later date, you must complete the entire LFC process to enable.

Load From Cold Procedure

- ❑ Shutdown the System
- ❑ Install Operating System
- ❑ Install CT Application Software
- ❑ Set Time and Date
- ❑ Start CT Application Software
- ❑ Restore System State
- ❑ Download Flash Firmware
- ❑ Adding System Options
- ❑ Install Cameras
- ❑ Configure User's System Access
- ❑ Shutdown System
- ❑ Verify System Operation

FMI 25338 Section 2
Direction 2362590-100

NOTE: Avoid disabling current customer Software Options!

Be sure to re-enable all previously enabled LightSpeed options using the on-site option DVD's or MOD's.

System & Console Setup Checks

- ❑ Verify Gantry Revolution Counter Restore

FMI 25338 Section 3

Load Class C & Class M Software

- ❑ Install Class C Software
- ❑ Install Class M Software

FMI 25338 Section 4

NOTE: Avoid Class M Software Partial Load Problems!

The 1st attempt to load Class M software may fail if a reboot was completed prior to the load.

Check usr/adm Class M software load log for loading errors to make sure the load completed successfully. If not, repeat the Class M Software load.

Re-Initialize Insite Connectivity

- ❑ Restore System State (Insite config data)
- ❑ Enable IIP Admin Config Utility
- ❑ Configure ProDiags Schedule
- ❑ Configure HealthPage
- ❑ Check Device Connection Setup
- ❑ Completing the Insite Re-Initialization
- ❑ Run ConfigLink
- ❑ Reboot System
- ❑ Save System State

FMI 25338 Section 5

Finishing Up

- ❑ Customer Documentation
- ❑ FMI Closure

FMI 25338 Section 6

Steering Guide for GRE (GOC3)

Prepare System For Upgrade

- ❑ Software Revision Check
- ❑ Archive Image Data
- ❑ System Operation Verification
- ❑ Check Software Loading Devices
- ❑ Save System State & CSA Protocols
- ❑ Record Key System Information
- ❑ Network Configuration Info
- ❑ HIS/RIS Configuration Info (Connect Pro)
- ❑ Insite Connectivity Info
- ❑ BroadBand Insite Connected Systems
- ❑ Installed Options
- ❑ Record Latest Gantry Revolution Entry

FMI 25338 Section 1
Direction 2378261-100
Chapter 3

Load From Cold Procedure

- ❑ LFC - CDROM OS Installation on HP Host Computer
- ❑ LFC - CDROM Application Software Installation on the HP Host Computer
- ❑ LFC - CDROM OS Installation on DARC
- ❑ LFC - CDROM DARC Application Installation on HP Host Computer
- ❑ Reboot the System
- ❑ Start Up the CT Applications
- ❑ Restore System to Operational Conditions
- ❑ Restore System State
- ❑ Retro Recon Test
- ❑ FLASH Download
- ❑ Restore System Options
- ❑ Configure User's System Access
- ❑ HIPAA Present set to ON - Admin Screen
- ❑ Set Time and Date
- ❑ Restart the System
- ❑ System Sanity Testing

FMI 25338 Section 2
Direction 2378261-100
Chapter 3

> **NOTE: Avoid re-loading OS software to enable other-than-English keyboard.**
>
> Be sure to select the correct language keyboard in the initial OS load process. This is the only method available to set the system to recognize an "other-than-English" keyboard. If you choose to install an international language keybboard at a later date, you must complete the entire LFC process to enable.

> **NOTE: Avoid disabling current customer Software Options!**
>
> Be sure to re-enable all previously enabled LightSpeed options using the on-site option DVD's or MOD's.

System & Console Setup Checks

- ❑ Verify Gantry Revolution Counter Restore
- ❑ Check Media Converter Switch Settings

FMI 25338 Section 3

Load Class C & Class M Software

- ❑ Install Class C Software
- ❑ Install Class M Software

FMI 25338 Section 4

> **NOTE: Avoid Class M Software Partial Load Problems!**
>
> The 1st attempt to load Class M software may fail if a reboot was completed prior to the load.
>
> Check usr/adm Class M software load log for loading errors to make sure the load completed successfully. If not, repeat the Class M Software load.

Re-Initialize Insite Connectivity

- ❑ Restore System State
- ❑ Enable IIP Admin Config Utility
- ❑ Configure ProDiags Schedule
- ❑ Configure HealthPage
- ❑ Check Device Connection Setup
- ❑ Completing the Insite Re-Initialization
- ❑ Run ConfigLink
- ❑ Reboot System
- ❑ Save System State

FMI 25338 Section 5

Finishing Up

- ❑ Customer Documentation
- ❑ FMI Closure

FMI 25338 Section 6

Procedure

This FMI includes the following procedures:

- Preparing the System for Software Upgrade
- Load From Cold

1 Prepare System for Software Upgrade

1.1 Software Revision Check

Check the current version of CT Application Software on the system using the showprods command.

1. Open a shell.
2. At the prompt enter the following command

   showprods | grep Helios_OC ENTER

3. Verify that the response to the showprods inquiry displays one of the following (Linux M3, GRE M3, GRE M3.1, & Linux/GRE M4, respectively):

```
I     Helios_OC_Product   06/23/2003
      LightSpeed ver 402L.1 rel 402L.1_H4.1M3

I     Helios_OC_Product   06/23/2003
      LightSpeed ver 403L.4a rel 403L.4a_H4.2GREM3

I     Helios_OC_Product   06/23/2003
      LightSpeed ver 404L.2 rel 404L.2_H4.2GREM3.1

I     Helios_OC_Product   06/23/2003
      LightSpeed ver 405L.5 rel 405L.5_H4.2M4
```

Note: Only LightSpeed 4.X H16 Systems currently using the revisions of CT Application Software listed above are compatible with this FMI.

1.2 Archive Image Data

The following procedures (Software Load From Cold) will destroy all image data on the system. Prior to starting the software load, make sure that the customer has reconstructed and archived all images before proceeding.

1.3 System Operation Verification

Setup and perform at least one axial scan to verify that the system operated properly before beginning the upgrade.

1.4 Check Software Loading Devices

Since the CDROM Drives on Linux systems are seldom used except for loading system software, it is advisable at this time to perform a quick test on the ability of these devices before you begin the LFC procedure.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.1.3 Remote Shell – Check Internal Network

Section 3.1.4 HP Computer CDROM – Mount and Unmount Process

Section 3.1.5 Peripheral Tower DVD - Mount and Unmount Process

Section 3.1.6 Peripheral Tower DVD – Initializing a DVD

Section 3.1.7 Peripheral Tower MOD - Mount and Unmount Process

1.5 Save System State & CSA Protocols

This section details the procedure for archiving the system critical files (System State) and adding to them the customer protocol settings that previously were not included in the System State archive process. This newly created System State will be used during the upgrade Load From Cold Procedure.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.2.1 Save System State

Section 3.2.2 CSA Protocol Retention

1.6 Record Key System Configuration Information

To ensure all system configuration information can be restored after the Load From Cold process, record the following parameters. Although the Save System State utility saves most of these parameters, recording this information provides a means to check that these parameters are properly restored after the Restore State Process.

1.7 Network Configuration Info (found in /usr/g/config INFO)

Check CT LightSpeed Network Settings as follows:

1. Open a shell.
2. At the prompt enter the following command

   `cd /usr/g/config` ENTER

   `more INFO` ENTER

From the usr/g/config INFO file, record the following network configuration parameters:

Parameter	Line Name in /usr/g/config INFO	Record Parameter Value	Notes
Hospital Name	HOSPITAL_NAME		Can have apostrophe with IIP 3.2
Hostname	GATEWAY_HOSTNAME		
System IP Address	GATEWAY_IP	___.___.___.___	
Broadcast IP Address	GATEWAY_BROADCAST	___.___.___.___	
Netmask IP Address	GATEWAY_NETMASK	___.___.___.___	
Default Gateway	GATEWAY_DEFAULT	___.___.___.___	May have been used as the Insite Default Gateway in previous BroadBand Insite Checkout.
NIS Server	NIS_SERVER	___.___.___.___	

1.8 HIS/RIS Configuration Info (found in /usr/g/config INFO)

NOTE: If you do not have HIS/RIS, you can skip this section.

Check CT LightSpeed NIS/RIS Settings as follows:

1. Open a shell.
2. At the prompt enter the following command

   `cd /usr/g/config` ENTER

   `more INFO` ENTER

From the usr/g/config INFO file, record the following HIS/RIS configuration parameters:

Parameter	Line Name in /usr/g/config INFO	Record Parameter Value	Notes
HIS/RIS Title	HRCTTITLE		
HI/RIS Port Number	HRPORTNUM		
HIS/RIS AE Title	HRAETITLE		
HIS/RIS IP Address	HRIPADDR	___.___.___.___	

1.9 Insite Connectivity Info (found in /usr/g/insite .insiteINFO)

NOTE: If you don't have Insite, you can skip this section.

1. Open a UNIX shell.
2. At the prompt enter the following command

`cd /usr/g/insite` ENTER

`more .insiteINFO` ENTER

From the usr/g/insite .insiteINFO file, record the following Insite Connectivity configuration parameters:

Parameter	**Line Name in /usr/g/insite .insiteINFO**	**Record Parameter Value**	**Notes**
Connection Type	INS_ConnectionType		
Modem Type	INS_ModemType		Modem Connectivity Only
Modem Phone Number	INS_PhoneNumber		Modem Connectivity Only
Modem Dial Prefix	INS_DialPrefix		Modem Connectivity Only
Modem Tone or Pulse Setting	INS_ToneOrPulse		Modem Connectivity Only
Modem Country	INS_Country		Modem Connectivity Only
Modem String	INS_ModemString		Modem Connectivity Only
Modem TTY Port	INS_TtyPort		Modem Connectivity Only
Modem TTY Speed	INS_TtySpeed		Modem Connectivity Only
BroadBand Insite Gateway IP	INS_GatewayIP	___.___.___.___	BroadBand Connectivity Only
BroadBand Gateway Mask	INS_GatewayMASK	___.___.___.___	BroadBand Connectivity Only

1.10 BroadBand Insite Connected Systems

If your system has already been Insite Checked Out as a Broadband system, you will need to re-enter the Insite Gateway IP Address into the IIP Admin Configuration utility when reconfiguring Insite after the Load From Cold process.

To ensure that you enter the correct IP Address of the Insite dedicated Gateway, confirm this IP Address with one of the following resources:

- Ask the Hospital Network Administrator or IT contact.
- In the United States, check for the IP Address at the following web site or by calling the regular checkout number (800-321-7937):

  http://gein2.med.ge.com/vpni/secure/index.jsp
- In Europe, contact the regular toll free number to reach the OLCE.

IP Address: ______________________________

Perform this investigation prior to arriving on site.

1.11 Installed Options (found using swokinstall -p)

Check CT LightSpeed 16 Options Settings as follows:

1) Open a UNIX shell.
2) At the prompt enter the following command

   swokinstall –p ENTER

From the swokinstall –p output, record which of the following LightSpeed 16 system options are installed on the system:

Line Name in output file of swokinstall -p	**Enabled (Y/N)**	**Permanent**	**Flex Trial**	**Standard LightSpeed 16 Option or Purchased Option**
Smart Prep				Standard
Smart Speed				Standard
Helical Tilt				Standard
Direct-3D				Standard
VariViewer				Standard
AutomA				Standard
Power440				Standard
Tube6_3MHU				Standard
3000 Image Series				Standard
Patient-16-slice				Standard
Thin Twin Helical				Standard
Connect Pro				Purchased Option
DentaScan				Purchased Option

Line Name in output file of swokinstall -p	Enabled (Y/N)	Permanent	Flex Trial	Standard LightSpeed 16 Option or Purchased Option
Smart Score				Purchased Option
Smart Score Pro				Purchased Option
CardIQ SnapShot				Purchased Option
Smart Step				Purchased Option
CTPerfusion2				Purchased Option
VolumeViewer				Purchased Option
AdvVesselAnalysis				Purchased Option
CTColono				Purchased Option
CardIQ2				Purchased Option

NOTE: Make sure that for each Option enabled, there is an on-site DVD that contains the key(s) for the Options presently enabled. Without the DVD keys, you will not be able to restore options on the system that were previously enabled.

1.12 Record Latest Gantry Revolution Counter Entry (found in getStats)

Record the latest system Gantry Revolution Counter entry as follows:

1. Open a shell.
2. Start the TTY getStats utility by typing the following at the ctuser prompt:

   `getStats` ENTER
3. Display the list of daily Gantry Revolution Stats by typing the following at the getStats "`Requested Command:`" line:

   `12` ENTER
4. Record the latest (list of last 30 days) Gantry Revolution Counter Value from the top of the list:

```
Requested Command: 12
  *****************************************************
  *          GANTRY Revolution Counter STATISTICS      *
  *****************************************************

Date: Tue Jan 27 13:33:22 2004  Total Gantry Revolution :  404773
```

   Total Gantry Revolutions: ____________________

5. Exit the getStats utility by typing the following at the "`Requested Command:`" line:

   `1` ENTER

2 Load From Cold Procedure

After system operation has been verified, all images have been archived a Save System State, and a record of key system configuration parameter has been completed, you are ready to upgrade the system software.

2.1 Shutdown Through Start CT Applications Steps

Linux/GRE Consoles

Using the LightSpeed GRE Software Installation Procedure document (2378261-100) and the software included in the FMI kit, perform a software load.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.3	Load From Cold Installation Procedures
Section 3.3.1	LFC – CDROM OS Installation on HP Host Computer
Section 3.3.2	LFC – CDROM Application Software Installation on the HP Host Computer
Section 3.3.3	LFC – CDROM OS Installation on DARC
Section 3.3.4	LFC – CDROM DARC Application Installation on HP Host Computer
Section 3.4	Reboot the System
Section 3.5	Start Up the CT Applications

Linux/Pegasus Consoles

Using the LightSpeed Linux Software Installation Procedure document (2362590-100) and the software included in the FMI kit, perform a software load.

Reference: Direction 2362590-100, Linux Software Installation

Section 3	Load Procedure
Section 3.1	Shutdown the System
Section 3.2	Install Operating System
Section 3.3	Install CT Application Software
Section 3.4	Set Time and Date
Section 3.5	Start CT Application Software

2.2 Restore System To Operational Conditions

Linux/GRE Consoles

Using the LightSpeed GRE Software Installation Procedure document (2378261-100) and the software included in the FMI kit, perform a software load.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.9	Restore System State (and CSA Protocols)
Section 3.10	Retro Recon Test
Section 3.11	FLASH Download

Linux/Pegasus Consoles

Using the LightSpeed Linux Software Installation Procedure document (2362590-100) and the software included in the FMI kit, perform a software load.

Reference: Direction 2362590-100, Linux Software Installation

Section 3.6	Restore System State
Section 3.7	Download Flash Firmware

NOTE: Make sure that the Key System Information recorded in Section 1 (Network Configuration Info and HIS/RIS Configuration Info) is correctly set as recorded. If not, use the following to:

- reconfig utility to reset Network Configuration Info.
- Connect Pro utility to reset the HIS/RIS Configuration Info.

2.3 Restore System Options

Using the record of installed options prior to the software load, install all the options using the DVD keys for each.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.12	Install Options

Reference: Direction 2362590-100, Linux Software Installation

Section 3.8	Adding Software Options
Section 3.9	Install Cameras

2.4 Complete the Load From Cold Process

Reference: Direction 2378261-100, GRE Software Installation

Section 3.6	Configure Users System Access
Section 3.7	HIPAA Present set to ON – Admin Screen
Section 3.13	Set Time and Date
Section 3.14	Restart the System
Section 3.15	System Sanity Scanning

Reference: Direction 2362590-100, Linux Software Installation

Section 3.10	Configure User's System Access
Section 3.11	Shutdown System
Section 3.12	Verify System Operation

NOTE: AVOID LOSING INSITE CONNECTIVITY CONFIGURATION DATA!!

DO NOT SAVE SYSTEM STATE UNTIL YOU'VE COMPLETED LOADING CLASS C AND CLASS M SOFTWARE AND RE-INTIALIZED INSITE CONNECTIVITY. PERFORMING THE SAVE STATE PROCESS PRIOR TO LOADING CLASS C AND CLASS M SOFTWARE MAY CAUSE THE LOSS OF CRITICAL INSITE CONFIGURATION PARAMETERS AND RESULT IN THE NEED TO PERFORM AN INSITE CHECKOUT TO RESTORE CONNECTIVITY.

3 System and Console Setup Checks

3.1 Verify Gantry Revolution Counter Restore

With the release of the Linux Console revision CT Application Software, a problem was found with the gantry revolution counter utility that greatly over calculated the actual gantry revolutions. With this release, the utility has been fixed but to make sure that the correct value has been restored, perform the following:

1. Open a shell.
2. Start the TTY getStats utility by typing the following at the ctuser prompt:

   `getStats` ENTER

3. Display the list of daily Gantry Revolution Stats by typing the following at the getStats "`Requested Command:`" line:

   `12` ENTER

4. Verify the latest Gantry Revolution Counter Value matches the value recorded earlier:

```
Requested Command: 12
  *****************************************************
  *          GANTRY Revolution Counter STATISTICS     *
  *****************************************************


Date: Tue Jan 27 13:33:22 2004  Total Gantry Revolution :
```

If the Total Gantry Revolution value does not match the value recorded earlier, perform the following:

1. Start the Set Gantry Rev Counter setup by typing the following at the "`Requested Command:`" line:

   `25` ENTER (Set Gantry Rev Counter Stats)

2. When prompted with "`Do you want to update stats displayed?`" type:

   `Yes` ENTER

3. When prompted with `Enter the No. of Revolutions`, type the value recorded earlier or an estimated value using the following:

   200,000 revolutions per month x number of months system has been installed.

   Example:

   For a system that has been installed for four months, type:

   `800000`

   The getStats utility responds with the following message:

   `Updated Gantry Rev Stats Successfully..`

4. When the Gantry Rev Stats have completed updating and displays the getStats menu, exit the getStats utility by typing:

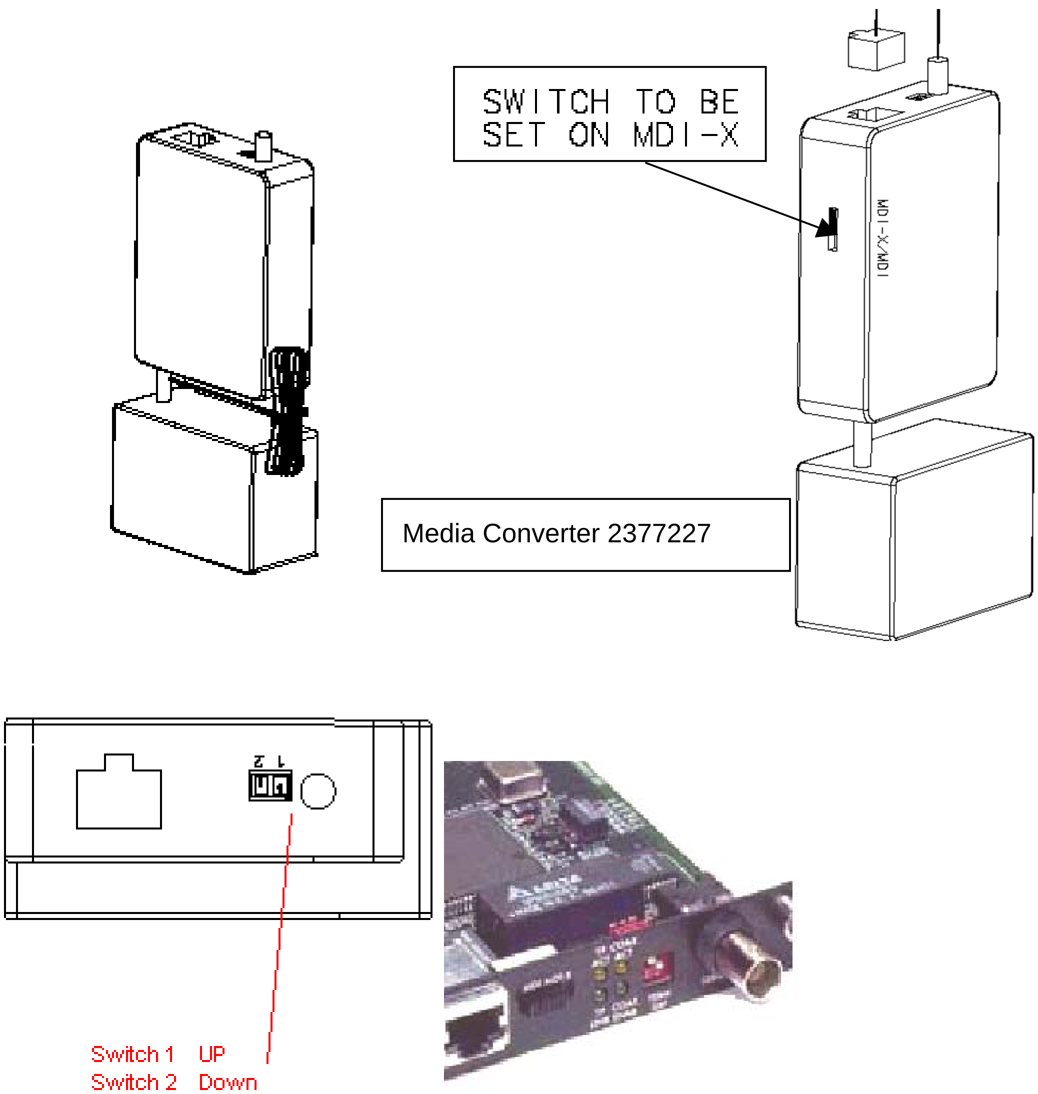
1 ENTER (exit)

3.2 Check Media Converter Switch Settings

Note: Only for Linux/GRE Consoles. With the intial release of the Linux/GRE Consoles, some consoles were shipped with the Media Converter Switch Setting improperly set. Check to make sure this setting is correct using Figure 1.

Figure 1
Checking Console Media Converter Switch Settings

Switch 1	UP (away from #)
Switch 2	Down (Toward #)



4 Load Class C and Class M Software

Notice: Avoid Partial Load of Class M Software

To ensure a proper load of the Class M Software, load Class C Software and immediately load Class M Software without rebooting the system between CD loads. The Class M Software CD load will fail to mount if not completed in this sequence. If the Class M Software install log exhibits Errors, usually a repeat attempt at loading Class M Software CD succeeds.

4.1 Load Class C Software

Load provided Class C software as follows:

1. Open a shell.
2. Enter root as superuser:
   a. At the ctuser prompt, type: **su -** ENTER
   b. At the Password prompt, type: **#bigguy** ENTER
3. Insert the Class C software CD into HP computer DVD Drive. Wait for the disk read to STOP.
4. Install the Class C software on the system. At the root prompt, type:

   install.classC ENTER
5. Close Class C install reporting shell.
6. Verify Class C is properly loaded on the system.
   a. At the root prompt, type: **cd /var/adm** ENTER
   b. At the root prompt, type: **ls** ENTER
   c. Find the install.classc.log file. If file found, this means the Class C software load completed.
   d. At the root prompt, type:

      more install.classc.log.<*yyyymmddhhmmss*>

      Refer to Appendix A for an example of a successful Class C software load.
7. Remove the Class C software CD from HP computer DVD Drive.

4.2 Load Class M Software

Load provided Class M software as follows:

1. Open a shell.
2. Enter root as superuser:
   a. At the ctuser prompt, type: **su –** ENTER
   b. At the Password prompt, type: **#bigguy** ENTER
3. Insert the Class M software CD into the HP computer DVD drive. Wait for the disk read to STOP.

4. Install the Class M software on the system. At the root prompt, type:

   `install.classM` ENTER

5. Verify Class M is properly loaded on the system.

   a. At the root prompt, type: **`cd /var/adm`** ENTER

   b. At the root prompt, type: **`ls`** ENTER

   c. Find the install.classM.log file. If file found, this means the Class M software load completed.

   d. At the root prompt, type:

      `more install.classM.log.<yyyymmddhhmmss>`

      Refer to Appendix A for an example of a successful Class M software load.

6. Remove the Class M software CD from HP computer DVD Drive.

5 Re-Initializing Insite Connectivity

This release of software contains a new IIP release (IIP 3.2-Linux). Changes in this IIP release are primarily in the following areas:

- Updates to support Multi-Tech modem firmware revision 8.19M
- Removal of IIP_DialPrefix ProDiags task
- Replacement of current iipHealthPage ProDiags task with a new Insite Data Mining task
- Auto Route to AutoSC in Level 5 setup

In addition, the Network Configuration Tab in the Insite Interactive Administration utility has been modified to eliminate the Use Default Gateway selection.

To ensure these IIP 3.2 features are correctly set, an Insite Checkout is NOT required. Because the basic Insite configuration information on the system should be correct from the original Insite Checkout, in most cases, it is only necessary to run ConfigLInk via exiting the IIP Admin Configuration utility Insite Checkout Tab Screen.

5.1 Restore System State To Retrieve Class M Configuration Parameters

Once Class M software has been loaded, you must repeat a Restore System State to recover all previous Insite and Class M configuration parameters.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.9 Restore System State

NOTE: Make sure that the Key System Information recorded in Section 1 (Insite Connectivity Info) is correctly set as recorded. If not, use the IIP Admin Configuration utility to reset Network Configuration Info as described in the following procedures.

5.2 Enable Insite Interactive Platform Configuration Utility

See Figure 2, Insite Interactive Platform Configuration Utility Gaining Access and Selecting ProDiags Utility.

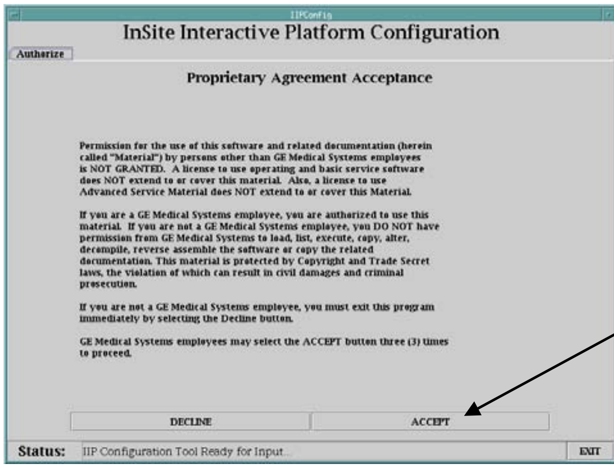
Enable the Insite Interactive Platform Configuration utility as follows:

1. Open a shell.
2. Enter root as superuser:
   a. At the ctuser prompt, type: **`su -`** ENTER
   b. At the Password prompt, type: **`#bigguy`** ENTER
3. Launch the IIP Admin Configuration utility. At the root prompt, type:

   `iipadmin config` ENTER

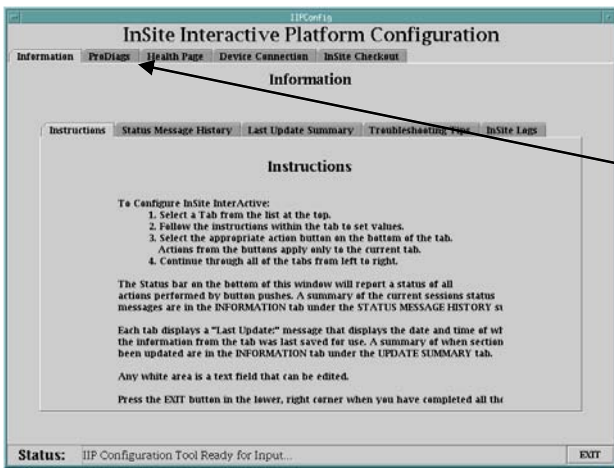
   The system displays the Insite Interactive Platform Configuration utility with the Proprietary Agreement Acceptance Screen.
4. Click on ACCEPT three times to display the Information Screen.
5. Click on the **ProDiags** Tab to display the ProDiags Screen.
6. Click on the **Custom** button to launch the independent ProDiags utility as shown in Figure 3.

Figure 2
Insite Interactive Platform Configuration Utility
Gaining Access and Selecting ProDiags Utility



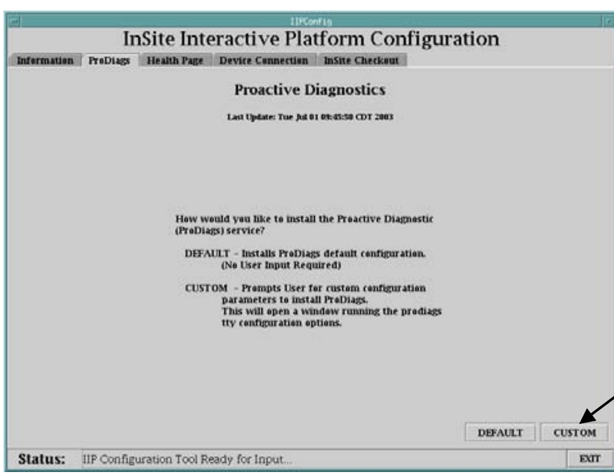
Proprietary Agreement Screen

1. Click **Accept** Three Times to gain access to the Insite Interactive Platform Configuration Utility



Information Screen

2. Click on the **ProDiags** Tab to display the IIP Admin Config Utility ProDiags Sceen



ProDiags Screen

3. Click on the **Custom** button to launch the independent ProDiags Utility

5.3 Configure ProDiags Schedule

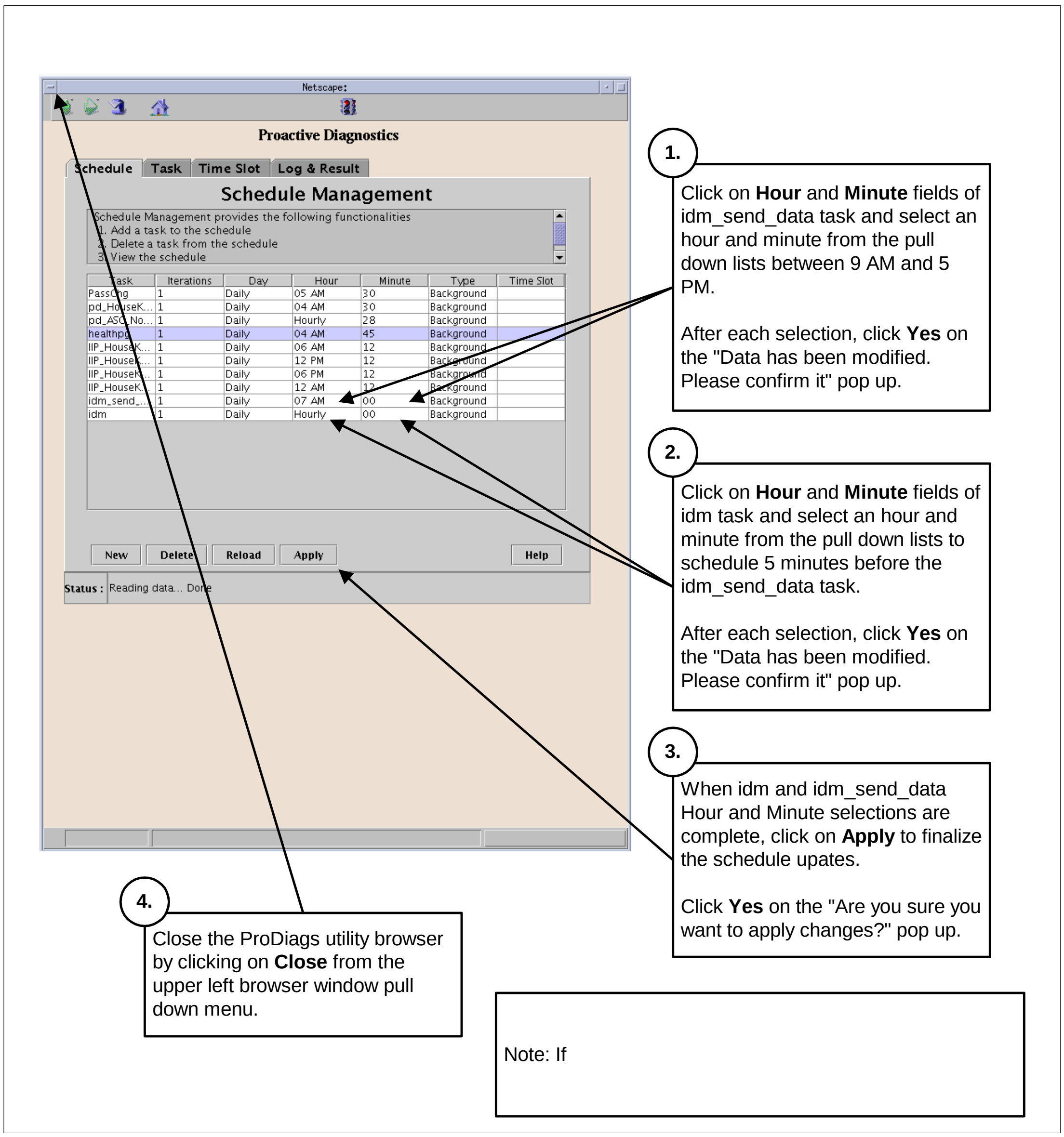
With this release of software, the IIP 3.2 package contains a new data collection package that pushes gesyslog data and a basic system configuration file to the On Line Center RDA Database. This IDM package contains two new ProDiags tasks (idm and idm_send_data) that must be configured to match the site's operational usage.

The default ProDiags schedule provided in this release should be adequate for most systems. However, we recommend that you review this schedule and update according to your particular site's normal operation.

Updating ProDiags Scheduled Tasks

See Figure 3 and refer to detailed steps on the following page.

Figure 3
ProDiags Graphical Utility
Schedule Tab – Modifying Scheduled Tasks



Use the following sequence to update the scheduled time (Hour and Minute) of the idm and idm_send_data tasks:

ProDiags idm_send_data task

- Click on the **idm_send_data** task **Hour** field.
- From the pull down list, select an hour that the system is normally on (typically between 09:00 and 17:00).
- On the "Data has been modified. Please confirm it" pop-up, click on YES.
- Click on the **idm_send_data** task **Minute** field.
- From the pull down list, select a minute (preferably an odd minute such as 3, 14, 22, etc).
- On the "Data has been modified. Please confirm it" pop-up, click on YES.

ProDiags idm task: Set up the idm task to run five minutes before the idm_send_data task.

- Click on the **idm** task **Hour** field.
- From the pull down list, select an hour that sets the idm task to run five minutes before the idm_send_data task.
- On the "Data has been modified. Please confirm it" pop-up, click on YES.
- Click on the **idm_send_data** task **Hour** field.
- From the pull down list, select a minute that sets the idm task to run five minutes before the idm_send_data task.
- On the "Data has been modified. Please confirm it" pop-up, click on YES.

When completed with the idm and idm_send_data modifications, click on the APPLY button at the bottom of the the Schedule Tab. On the "Apply change confirmation" pop-up, click on YES.

NOTICE: **If the schedule change failed due to a permissions denied error, execute the following change-mode command to make the file named bg_schedule_data read/writable:**

- With the ProDiags GUI still open, open another shell
- Become root user:
  type: su - <Enter>
  type: #bigguy <Enter> at the Password prompt
- Set Permissions for bg_schedule_data:
  type: cd /usr/g/insite/ProDiags.schedule
  type: chmod 777 bg_schedule_data

Repeat the Apply click on the ProDiags Schedule Tab. The update should occur without failures.

Close the ProDiags utility browser by selecting **Close** from the upper left browser window pull down menu. The ProDiags browser should close and the Insite Interactive Platform Configuration utility should display the HealthPage Tab as shown in Figure 4.

5.4 Configure HealthPage

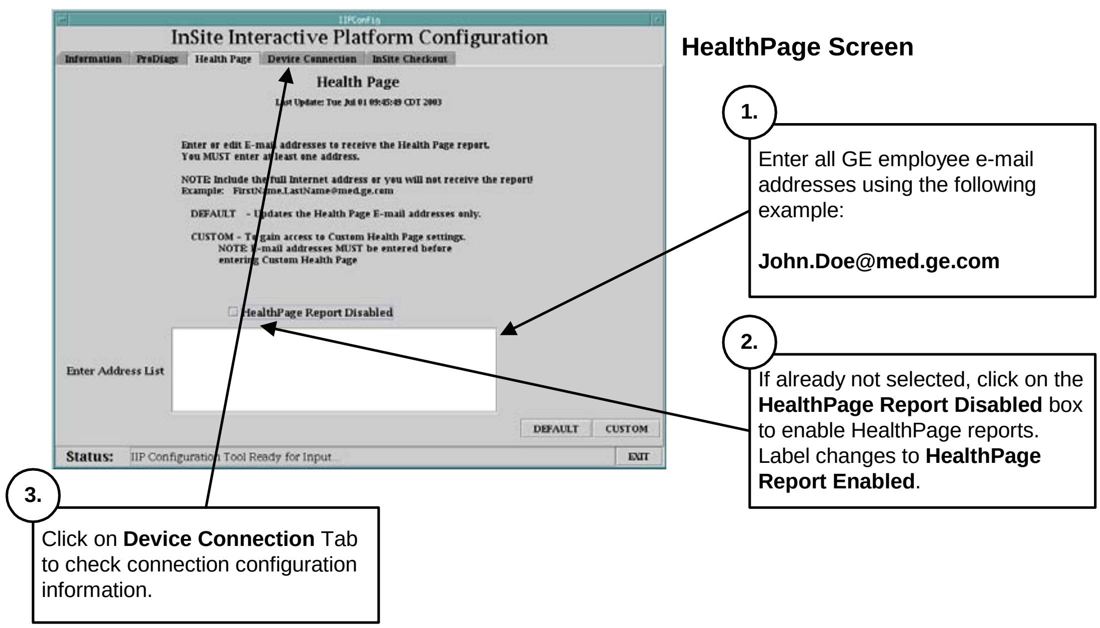
After completing the ProDiags setup, the IIP Admin Configuration utility automatically switches to the HealthPage Tab as shown in Figure 4.

The Restore State process (after loading Class M software) restored the following information for the HealthPage Reports:

- All e-mail addresses of targeted recipients that were previously loaded
- HealthPage report contents that was previously configured
- HealthPage report schedule that was previously configured

Unless you have recent changes to any of these three items, you can skip the HealthPage Tab of the IIP Admin Configuration utility and select the Device Connection Tab. If you have changes that you want to make (e-mail addresses, report content, report schedule), follow the instructions on the IIP Admin Configuration utility.

Figure 4
Insite Interactive Platform Configuration Utility
HealthPage Tab



5.5 Check Device Connection Setup

5.5.1 Modem Setup

When selecting the Device Connection Tab, the IIP Admin Configuration utility automatically displays the Modem Setup Screen as shown in Figure 5.

If your system was configured for Modem connectivity, the Restore State process (after loading Class M software) restored all the Modem setup parameters. To verify, refer to Section 1 of this FMI where you recorded Insite Connectivity Info.

- If there were no Modem parameters in the .insiteINFO file but there were BroadBand parameters, skip to the Check BroadBand Device Connection Setup procedure.

- If there were Modem parameters in the .insiteINFO file but the parameters displayed in the Modem Setup Screen do not match what you previously recorded, enter the correct values as required.

NOTE: The Modem configuration parameters should be pre-loaded from the Restore State process. Do not change settings unless these values were not properly restored.

If required, follow the instructions in Figure 5. The default values are usually the correct values or most commonly used. Parameter selections that do not require customization are:

CPU Serial Port Name

Figure 5
Insite Interactive Platform Configuration Utility
Device Connection Tab – Modem Setup

1.

Select the Dial-out Prefix value recorded from the previous .insiteINFO file (INS_DialPrefix) from the pull down menu. This value is the number you must use to dial out of the hospital facility.

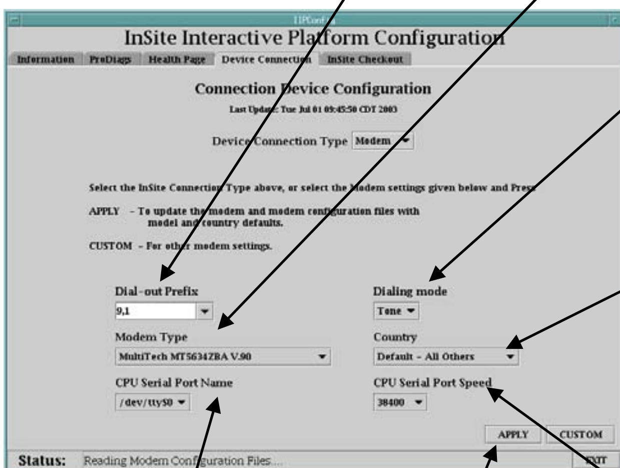
2.

Select the Modem Type value recorded from the previous .insiteINFO file (INS_ModemType) from the pull down menu.

3.

Select the Dialing Mode value recorded from the previous .insiteINFO file (INS_ToneOrPulse) from the pull down menu.

Device Connection Modem Screen



4.

Select the Country value recorded from the previous .insiteINFO file (INS_Country) from the pull down menu.

5.

Select the CPU Serial Port Speed value recorded from the previous .insiteINFO file (INS_TtySpeed) from the pull down menu.

We recommend that you select the highest Port Speed selection available (**115200**) regardless of the previous selection. Making this selection helps modem connected systems maximize connectivity performance.

6.

Select the CPU Serial Port Name value recorded from the previous .insiteINFO file (INS_TtyPort) from the pull down menu.

7.

Click on **Apply** to set configuration parameters and display the Insite Checkout Tab.

5.5.2 BroadBand Setup

If your system was previously BroadBand Insite connected, click on the Network button of the IIP Admin Configuration utility Device Connection Tab to display the Network Setup Screen as shown in Figure 6.

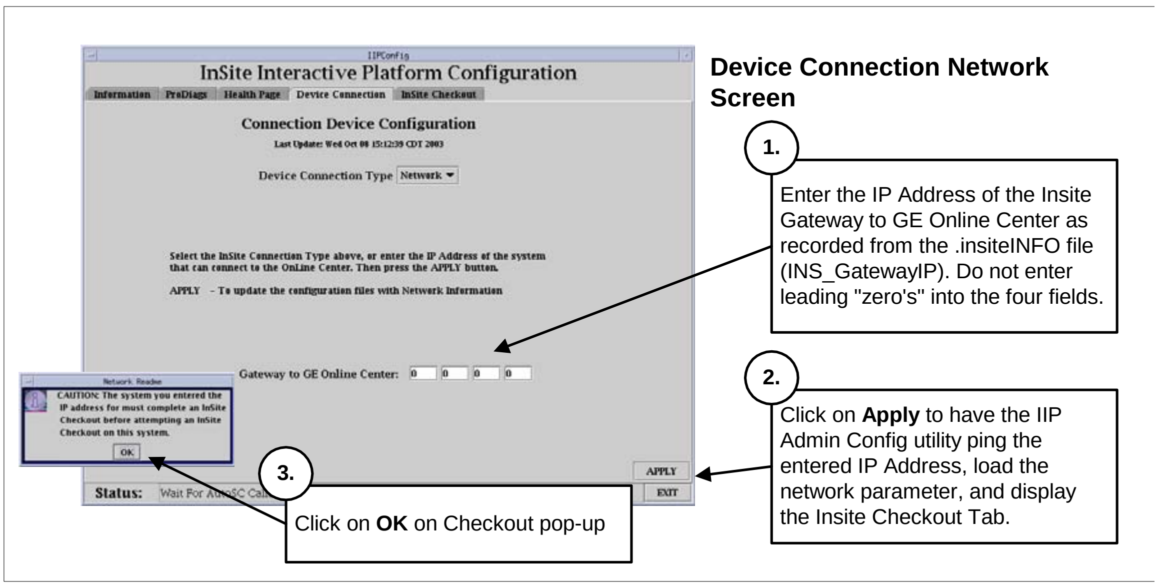
If your system was configured for BroadBand connectivity, the Restore State process (after loading Class M software) restored all the BroadBand setup parameters. To verify, refer to Section 1 of this FMI where you recorded Insite Connectivity Info.

NOTE: The BroadBand Network configuration parameters should be pre-loaded from the Restore State process. However, the on-site FE who performed the original Insite Checkout of the system may have used the hospital's imaging LAN default gateway IP Address (selection was available in the previous release of IIP).

DO NOT USE THE HOSPITAL WORKING LAN DEFAULT GATEWAY IP ADDRESS AS THE INSITE GATEWAY IP ADDRESS OTHER THAN AS A TEMPORARY MEANS TO CONNECT TO THE ON LINE CENTER!!

To find the correct Insite Gateway IP Address, use the contacts listed in Section 1.10, BroadBand Insite Connected Systems.

Figure 6
Insite Interactive Platform Configuration Utility
Device Connection Tab – Network Setup

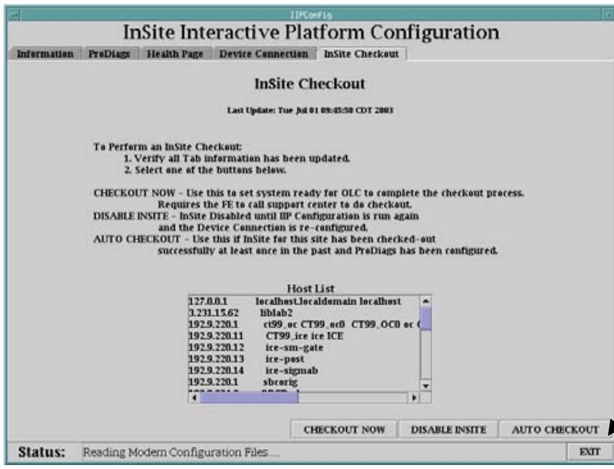


5.6 Completing the Insite Re-Initialization

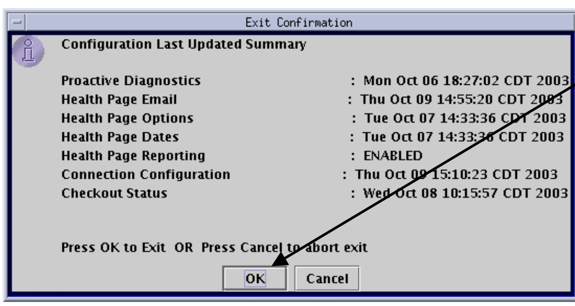
Follow the instructions provided in Figure 7 to exit the IIP Admin Configuration Utility.

Figure 7
Insite Interactive Platform Configuration Utility
Insite Checkout Tab – Exiting the IIP Admin Configuration Utility

Insite Checkout Screen



1. Click on the Exit button of the IIP Admin Config utility to exit the utility.



2. Click on **OK** on the Exit Confirmation pop-up to exit the IIP Admin Config utility and run the ConfigLink utility.

3. After the shell displays the root prompt, reboot the system to load Insite connectivity parameters. As the system is coming up, you should see the iLinq icon (if the original Checkout included iLinq) as a selectable desktop on the right console monitor.

5.7 Run ConfigLink

Run the Insite **ConfigLink** script to populate all configuration information required to fully restore connectivity parameters on the system.

1. Open a shell.
2. Enter root as superuser:
    a. At the ctuser prompt, type: **`su –`** ENTER
    b. At the Password prompt, type: **`#bigguy`** ENTER
3. Change directory to Insite. At the root prompt, type:

    `cd /usr/g/insite` ENTER
4. Run the ConfigLink script. At the root prompt, type:

    `ConfigLink` ENTER

Wait for the shell to return the root prompt to ensure the **ConfigLInk** process completes. When complete, exit root and close the shell.

5.8 Reboot System

To ensure all Insite configuration parameters are properly loaded into the operating system (route tables, etc.), perform a system reboot.

1. Click on the pink Shutdown button.
2. From the Shutdown selection pop-up, click on RESTART, then OK, to reboot the system.

5.9 Save System State

At the completion of the Load From Cold Process and Insite Re-Initialization process, ensure all system state parameters are saved using the Save System State Utility.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.2.1 Save System State

Reference: Direction 236590-100, Linux/Pegasus Software Installation

Section 2.3 Save System State

6 Finishing Up

6.1 Customer Documentation & Software

Make sure that the customer receives the Class A Operator/Service documentation and all the Software CD-ROM's provided with this FMI.

6.2 FMI Closure

Close FMI per regional procedure. Note that the FMI cannot be closed until the Product Locator cards are completed, sent in, and GIB is updated with all tracked items in this FMI.

Remove this FMI document from the customer site.

Appendix A - Class C Install Log File Example

A.1 Successful Class C Install Log File Example

[insite@ct01 ~]$ cd /var/adm

[insite@ct01 adm]$ more install.classc.log.20040219203324

ClassC CD is mounted

Detecting Configuration...

Loading LightSpeed Class C software ...

Loading //mnt/cdrom/Helios_CLASSC_Product-grem4-13_lnx.i386.rpm . . .

Loading //mnt/cdrom/Helios_H2_O2LESS_CLASSC_Product-grem4-13_lnx.i386.rpm .

Loading //mnt/cdrom/Helios_H2_PEGASUS_CLASSC_Product-grem4-13_lnx.i386.rpm

Loading //mnt/cdrom/GRE_CLASSC_Product-grem4-13_lnx.i386.rpm . . .

Finished loading HiSpeed CT/i Class C software

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagAxialFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagAxialFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagColFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagColFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagCommsFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagCommsFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagConsPBFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagConsPBFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagCradleFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagCradleFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagDasSerialFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagDasSerialFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagElevFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagElevFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagEtcBld

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagEtcBld

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagFilamentFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagFilamentFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagHelpScreen

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagHelpScreen

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagObcBld

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagObcBld

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagRpBld

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagRpBld

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagRpDp

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagRpDp

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagRpFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagRpFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagStcBld

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagStcBld

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagTableStatFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagTableStatFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/DiagTiltFunc

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/DiagTiltFunc

Checking for /usr/g/ctuser/app-defaults/Xapplresdir/History_viewer

Octane2: Modifying /usr/g/ctuser/app-defaults/Xapplresdir/History_viewer

Installation of Class C complete ...

[insite@ct01 adm]$

A.2 Successful Class M Install Log File Example

```
[insite@ct01 adm]$ more install.classM.log.20040125175849
ClassC CD is mounted
Load from CD
Installing GEMS class M package for InSite Interactive:
   /mnt/cdrom/classM-insite-3.2-Linux.tar.Z
...Killing ppp Processes...
OK
Loading /mnt/cdrom/Helios_CLASSMOC_Product-grem4-13_lnx.i386.rpm . . .
Patch 302003 has been installed successfully
Patch 302005 has been installed successfully
Patch 302006 has been installed successfully
Patch 302007 has been installed successfully
Patch 302010 has been installed successfully
Patch 302011 has been installed successfully
Patch 302015 has been installed successfully
Patch 302017 has been installed successfully
guard_window_start
guard_window_start.dat
guard_window_stop
miia.conf.local
rcocStart
RCOC.tar
rcoc_install
RCOC_20030513.tar.gz
RCOC has been installed successfully at /usr/g/insite/RCOC
info - Making a link for prodiags ...
info - updating bg_schedule_data and configuring idm task to run at 7 a.m.
info - Completed loading IIP system health page.

Unmounting mpoint : /mnt/cdrom
Class-M Installation complete.
[insite@ct01 adm]$
```